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SCRIP PROJECT:- Cyber Attack Investigation using Network Packet
Forensics

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“BSC.(Information Technology)”



“SCHOOL OF COMPUTER APPLICATION”

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Cyber Attack Investigation using Network Packet Forensics

Forensic Investigation Report

1. Introduction

In today's digital era, almost every online activity depends on network communication. As technology continues to grow, cyber threats such as hacking, phishing, malware, and unauthorized access are becoming more common. This makes network security and digital investigations more important than ever.

Network packet forensics is the process of capturing and analyzing network packets to understand how devices communicate and to investigate suspicious activities. Using Wireshark, every packet can be examined to identify important details such as IP addresses, ports, protocols, timestamps, and communication patterns. These details help investigators collect digital evidence and understand what happened during network communication.

The main objective of this project is to gain practical knowledge of network packet analysis using Wireshark. Live network traffic was captured and analyzed using different filters, including DNS, TCP, UDP, ICMP, TLS, and HTTP. This project helped me understand how data travels across a network, how different protocols work, and how packet-level analysis is used during a forensic investigation.

The knowledge gained from this project is valuable for careers in cybersecurity, digital forensics, incident response, and Security Operations Centers (SOC). In the future, packet analysis can be combined with Artificial Intelligence, Intrusion Detection Systems (IDS), and Security Information and Event Management (SIEM) platforms to improve real-time threat detection and strengthen cyber defense for organizations and future digital infrastructures.

2. Objectives

The primary objective of this project is to understand the process of network packet forensics using Wireshark and gain practical knowledge of network traffic analysis. The project was carried out with the following objectives:

Capture Live Network Traffic

- Capture real-time network packets generated during normal internet activities using Wireshark.
- Observe how devices communicate over a network and record the captured data for further analysis.

Analyze Network Protocols

- Study the behavior of commonly used protocols such as **DNS, TCP, UDP, ICMP, TLS, and HTTP**.
- Understand how each protocol functions and contributes to network communication.

Identify Suspicious Network Activity

- Examine packet details such as IP addresses, ports, protocols, timestamps, and TCP flags.
- Learn the methodology used to recognize unusual communication patterns that may indicate potential security threats.

Document Digital Evidence

- Collect packet captures and screenshots as digital evidence.
- Record observations, analysis, and findings in a structured forensic investigation report.

Develop Practical Cybersecurity Skills

- Gain hands-on experience in packet analysis and network forensics.
- Build a strong foundation for future roles in **Cybersecurity, Digital Forensics, Incident Response, and Security Operations Centers (SOC)**.

3. Investigation Methodology

The investigation methodology followed in this project was designed to understand how network traffic can be captured, analyzed, and documented using Wireshark. Each step was performed systematically to observe normal network communication and gain practical experience in packet-level analysis.



Step 1: Capturing Live Network Traffic

Wireshark was installed and configured to monitor the active Wi-Fi network interface. Live network traffic was generated by browsing websites, searching on Google, accessing online services, and using the **ping** command to create ICMP packets. This allowed different types of network communication to be captured in real time.



Step 2: Saving the Packet Capture

After capturing sufficient network traffic, the captured packets were saved in **PCAPNG** format. This file served as the primary digital evidence for the investigation and could be reopened later for detailed analysis without capturing the traffic again.



Step 3: Applying Display Filters

To simplify the analysis, Wireshark display filters were applied one by one. Filters such as **DNS, TCP, UDP, ICMP, TLS, HTTP, and `tcp.flags.syn==1 && tcp.flags.ack==0`** were used to isolate specific protocols and understand different types of network communication. This made it easier to study individual packet behavior instead of analyzing all captured packets at once.



Step 4: Analyzing Packet Information

Each filtered packet was carefully examined by observing its **Source IP Address, Destination IP Address, Source and Destination Ports, Protocol, Packet Length, Time, and Information** fields. TCP flags such as **SYN** and **ACK** were also analyzed to understand how reliable network connections are established and maintained. This helped distinguish normal communication from potentially unusual behavior.

Step 5: Understanding the Investigation Process

Although the live traffic mainly contained normal network communication, the project also focused on learning the methodology used in network forensic investigations. By studying packet patterns, retransmissions, and protocol behavior, I gained an understanding of how cybersecurity professionals investigate suspicious traffic, identify abnormal communication, and collect packet-level evidence during real-world security incidents.

Step 6: Documentation and Reporting

Finally, all observations, packet analysis, screenshots, and conclusions were documented in a structured forensic investigation report. The report summarizes the complete investigation process, explains the findings, and demonstrates how Wireshark can be used as an effective tool for network packet analysis and digital forensic investigations.

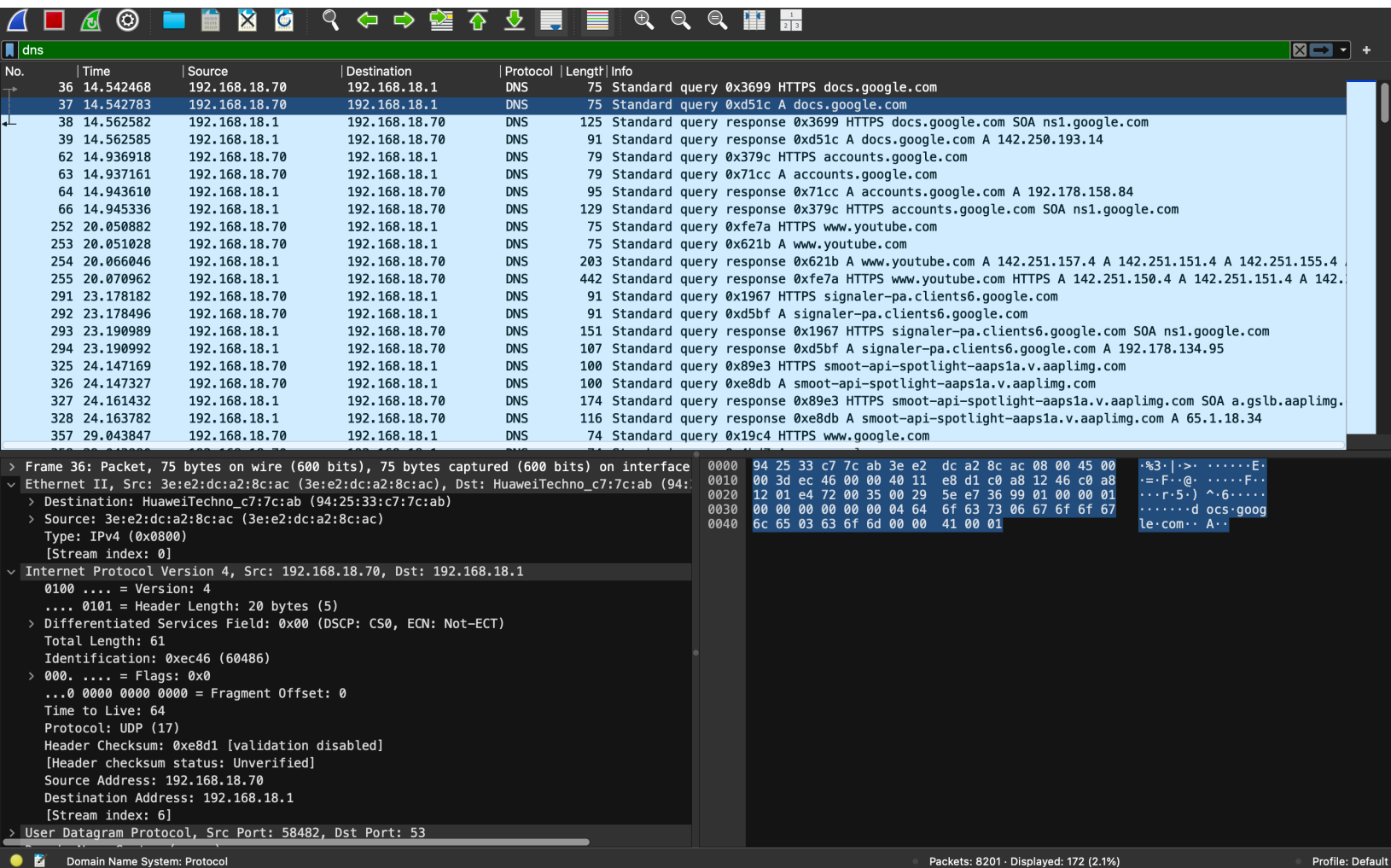
4. Packet Analysis

Packet analysis is a fundamental step in network forensics because it helps investigators understand how devices communicate and exchange data over a network. Every online activity, such as browsing websites, searching on Google, or sending a request to a server, generates network packets that contain valuable information, including source and destination IP addresses, port numbers, protocols, timestamps, packet lengths, and communication details. By examining these packets, it becomes possible to understand network behavior, verify whether communication is functioning normally, and gather digital evidence for forensic investigations.

In this project, Wireshark was used to capture and analyze live network traffic generated during normal internet usage. Different display filters such as **DNS, TCP, UDP, ICMP, TLS, HTTP, and TCP SYN** were applied to study individual protocols and their roles in network communication. The analysis provided practical knowledge of how data travels across a network, how reliable connections are established, how encrypted communication is secured, and how packet-level information can be used to investigate suspicious activities. The following sections present the observations and findings obtained from analyzing each protocol individually.

DNS (Domain Name System) Analysis

The **DNS** filter was used to observe how domain names are translated into IP addresses before communication begins. The captured packets showed DNS query and response messages whenever websites were accessed. This analysis helped in understanding the first step of network communication and verified that domain name resolution was functioning normally.



The image displays a Wireshark network traffic analysis interface. The top pane shows a list of captured packets, with the DNS filter applied. The middle pane shows the details of the selected packet (Frame 36), including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The bottom pane shows the raw packet data in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
36	14.542468	192.168.18.70	192.168.18.1	DNS	75	Standard query 0x3699 HTTPS docs.google.com
37	14.542783	192.168.18.70	192.168.18.1	DNS	75	Standard query 0xd51c A docs.google.com
38	14.562582	192.168.18.1	192.168.18.70	DNS	125	Standard query response 0x3699 HTTPS docs.google.com SOA ns1.google.com
39	14.562585	192.168.18.1	192.168.18.70	DNS	91	Standard query response 0xd51c A docs.google.com A 142.250.193.14
62	14.936918	192.168.18.70	192.168.18.1	DNS	79	Standard query 0x379c HTTPS accounts.google.com
63	14.937161	192.168.18.70	192.168.18.1	DNS	79	Standard query 0x71cc A accounts.google.com
64	14.943610	192.168.18.1	192.168.18.70	DNS	95	Standard query response 0x71cc A accounts.google.com A 192.178.158.84
66	14.945336	192.168.18.1	192.168.18.70	DNS	129	Standard query response 0x379c HTTPS accounts.google.com SOA ns1.google.com
252	20.050882	192.168.18.70	192.168.18.1	DNS	75	Standard query 0xfe7a HTTPS www.youtube.com
253	20.051028	192.168.18.70	192.168.18.1	DNS	75	Standard query 0x621b A www.youtube.com
254	20.066046	192.168.18.1	192.168.18.70	DNS	203	Standard query response 0x621b A www.youtube.com A 142.251.157.4 A 142.251.151.4 A 142.251.155.4
255	20.070962	192.168.18.1	192.168.18.70	DNS	442	Standard query response 0xfe7a HTTPS www.youtube.com HTTPS A 142.251.150.4 A 142.251.151.4 A 142.
291	23.178182	192.168.18.70	192.168.18.1	DNS	91	Standard query 0x1967 HTTPS signaler-pa.clients6.google.com
292	23.178496	192.168.18.70	192.168.18.1	DNS	91	Standard query 0xd5bf A signaler-pa.clients6.google.com
293	23.190989	192.168.18.1	192.168.18.70	DNS	151	Standard query response 0x1967 HTTPS signaler-pa.clients6.google.com SOA ns1.google.com
294	23.190992	192.168.18.1	192.168.18.70	DNS	107	Standard query response 0xd5bf A signaler-pa.clients6.google.com A 192.178.134.95
325	24.147169	192.168.18.70	192.168.18.1	DNS	100	Standard query 0x89e3 HTTPS smoot-api-spotlight-aaps1a.v.aaplimg.com
326	24.147327	192.168.18.70	192.168.18.1	DNS	100	Standard query 0xe8db A smoot-api-spotlight-aaps1a.v.aaplimg.com
327	24.161432	192.168.18.1	192.168.18.70	DNS	174	Standard query response 0x89e3 HTTPS smoot-api-spotlight-aaps1a.v.aaplimg.com SOA a.gslb.aaplimg.
328	24.163782	192.168.18.1	192.168.18.70	DNS	116	Standard query response 0xe8db A smoot-api-spotlight-aaps1a.v.aaplimg.com A 65.1.18.34
357	29.043847	192.168.18.70	192.168.18.1	DNS	74	Standard query 0x19c4 HTTPS www.google.com

Frame 36: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface
> Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
> Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
> Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)
Type: IPv4 (0x0800)
[Stream index: 0]
> Internet Protocol Version 4, Src: 192.168.18.70, Dst: 192.168.18.1
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 61
Identification: 0xec46 (60486)
> 000. = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xe8d1 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.18.70
Destination Address: 192.168.18.1
[Stream index: 6]
> User Datagram Protocol, Src Port: 58482, Dst Port: 53

Domain Name System: Protocol

Packets: 8201 · Displayed: 172 (2.1%)

Profile: Default

DNS ANALYSIS



TCP (Transmission Control Protocol) Analysis

The **TCP** filter was applied to study reliable communication between the client and the server. The captured packets showed the TCP three-way handshake using **SYN**, **SYN-ACK**, and **ACK** flags before data transmission. This confirmed that connections were established successfully and demonstrated how TCP ensures reliable data delivery.

The screenshot displays a Wireshark packet capture analysis of TCP traffic. The main pane shows a list of 23 packets. Packet 6 is selected, showing details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet details pane shows the TCP header and options. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.18.70	104.18.32.47	TCP	1454	51531 → 443 [ACK] Seq=1 Ack=1 Win=18748 Len=1388 TSval=2449598385 TSecr=1761781808 [TCP PDU reass]
2	0.000004	192.168.18.70	104.18.32.47	TCP	1454	51531 → 443 [ACK] Seq=1389 Ack=1 Win=18748 Len=1388 TSval=2449598385 TSecr=1761781808 [TCP PDU re]
3	0.000006	192.168.18.70	104.18.32.47	TCP	1454	51531 → 443 [ACK] Seq=2777 Ack=1 Win=18748 Len=1388 TSval=2449598385 TSecr=1761781808 [TCP PDU re]
4	0.000006	192.168.18.70	104.18.32.47	TCP	1454	51531 → 443 [ACK] Seq=4165 Ack=1 Win=18748 Len=1388 TSval=2449598385 TSecr=1761781808 [TCP PDU re]
5	0.000007	192.168.18.70	104.18.32.47	TCP	1454	51531 → 443 [ACK] Seq=5553 Ack=1 Win=18748 Len=1388 TSval=2449598385 TSecr=1761781808 [TCP PDU re]
6	0.000086	192.168.18.70	104.18.32.47	TLSv1...	624	Application Data
7	0.000134	192.168.18.70	104.18.32.47	TLSv1...	1201	Application Data
8	0.020879	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=1389 Win=84 Len=0 TSval=1761785946 TSecr=2449598385
9	0.020912	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=2777 Win=84 Len=0 TSval=1761785946 TSecr=2449598385
10	0.020913	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=5553 Win=84 Len=0 TSval=1761785946 TSecr=2449598385
11	0.020914	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=6941 Win=84 Len=0 TSval=1761785946 TSecr=2449598385
12	0.020915	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=7499 Win=84 Len=0 TSval=1761785946 TSecr=2449598385
13	0.020916	104.18.32.47	192.168.18.70	TCP	66	443 → 51531 [ACK] Seq=1 Ack=8634 Win=84 Len=0 TSval=1761785946 TSecr=2449598386
14	0.070953	104.18.32.47	192.168.18.70	TLSv1...	540	Application Data
15	0.070955	104.18.32.47	192.168.18.70	TLSv1...	144	Application Data, Application Data
16	0.071153	192.168.18.70	104.18.32.47	TCP	66	51531 → 443 [ACK] Seq=8634 Ack=553 Win=18740 Len=0 TSval=2449598457 TSecr=1761786000
19	4.648304	192.168.18.70	35.190.80.1	TCP	54	51409 → 443 [ACK] Seq=1 Ack=1 Win=2048 Len=0
20	4.678710	35.190.80.1	192.168.18.70	TCP	66	[TCP ACKed unseen segment] 443 → 51409 [ACK] Seq=1 Ack=2 Win=1031 Len=0 TSval=4144197860 TSecr=42
21	9.093107	172.67.72.108	192.168.18.70	TLSv1...	91	Application Data
22	9.093304	192.168.18.70	172.67.72.108	TCP	66	51540 → 443 [ACK] Seq=1 Ack=26 Win=2048 Len=0 TSval=4275103760 TSecr=4006990915
23	9.094778	192.168.18.70	172.67.72.108	TLSv1...	95	Application Data

Frame 6: Packet, 624 bytes on wire (4992 bits), 624 bytes captured (4992 bits) on interf. 0000

Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)

Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)

Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.18.70, Dst: 104.18.32.47

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x02 (DSCP: CS0, ECN: ECT(0))

Total Length: 610

Identification: 0x0000 (0)

> 010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: TCP (6)

Header Checksum: 0xdd64 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.18.70

Destination Address: 104.18.32.47

[Stream index: 0]

Transmission Control Protocol, Src Port: 51531, Dst Port: 443, Seq: 6941, Ack: 1, Len: 5

Packet (624 bytes) Reassembled TCP (7498 bytes)

Packets: 8201 - Displayed: 3845 (46.9%) Profile: Default

TCP ANALYSIS



ICMP (Internet Control Message Protocol) Analysis

The **ICMP** filter was used after generating network traffic with the **ping** command. Echo Request and Echo Reply packets were successfully captured, confirming that the destination host was reachable. This protocol is commonly used for network testing and troubleshooting.

The image shows a Wireshark packet capture analysis of ICMP Echo (ping) traffic. The packet list on the left shows several ICMP Echo (ping) requests and replies between 192.168.18.70 and 192.168.18.1. The selected packet (No. 1364) is an ICMP Echo (ping) request from 192.168.18.70 to 192.168.18.1. The packet details on the right show the Ethernet II header, Internet Protocol Version 4 header, and the Internet Control Message Protocol (ICMP) header. The ICMP header shows the type as Echo (ping) request, id as 0x9fb5, seq as 0/0, and ttl as 64. The packet bytes on the right show the raw data of the ICMP Echo request.

No.	Time	Source	Destination	Protocol	Length	Info
1364	46.030079	192.168.18.70	192.168.18.1	ICMP	70	Destination unreachable (Port unreachable)
7117	68.073670	192.168.18.70	142.250.182.110	ICMP	98	Echo (ping) request id=0x9fb5, seq=0/0, ttl=64 (reply in 7118)
7118	68.090766	142.250.182.110	192.168.18.70	ICMP	98	Echo (ping) reply id=0x9fb5, seq=0/0, ttl=117 (request in 7117)
7119	69.078267	192.168.18.70	142.250.182.110	ICMP	98	Echo (ping) request id=0x9fb5, seq=1/256, ttl=64 (reply in 7120)
7120	69.104118	142.250.182.110	192.168.18.70	ICMP	98	Echo (ping) reply id=0x9fb5, seq=1/256, ttl=117 (request in 7119)
7125	70.081290	192.168.18.70	142.250.182.110	ICMP	98	Echo (ping) request id=0x9fb5, seq=2/512, ttl=64 (reply in 7126)
7126	70.106484	142.250.182.110	192.168.18.70	ICMP	98	Echo (ping) reply id=0x9fb5, seq=2/512, ttl=117 (request in 7125)
7129	71.086609	192.168.18.70	142.250.182.110	ICMP	98	Echo (ping) request id=0x9fb5, seq=3/768, ttl=64 (reply in 7130)
7130	71.111611	142.250.182.110	192.168.18.70	ICMP	98	Echo (ping) reply id=0x9fb5, seq=3/768, ttl=117 (request in 7129)

Frame 1364: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface...

Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)

Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)

Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.18.70, Dst: 192.168.18.1

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 56

Identification: 0xd6ef (55023)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0xfe3d [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.18.70

Destination Address: 192.168.18.1

[Stream index: 6]

Internet Control Message Protocol

Internet Control Message Protocol: Protocol

Packets: 8201 · Displayed: 9 (0.1%)

Profile: Default

ICMP ANALYSIS

TLS (Transport Layer Security) Analysis

The TLS filter showed encrypted communication between the browser and secure websites. Since most modern websites use HTTPS, the captured traffic appeared as TLS packets instead of HTTP. This demonstrated how encryption protects sensitive information during data transmission.

The image displays a Wireshark network traffic capture. The top pane shows a list of packets, with packet 27 selected. The middle pane shows the details of the selected packet, which is a TLSv1.2 packet. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packets List:

No.	Time	Source	Destination	Protocol	Length	Info
6	0.000086	192.168.18.70	104.18.32.47	TLSv1...	624	Application Data
7	0.000134	192.168.18.70	104.18.32.47	TLSv1...	1201	Application Data
14	0.070953	104.18.32.47	192.168.18.70	TLSv1...	540	Application Data
15	0.070955	104.18.32.47	192.168.18.70	TLSv1...	144	Application Data, Application Data
21	0.093107	172.67.72.108	192.168.18.70	TLSv1...	91	Application Data
23	0.094778	192.168.18.70	172.67.72.108	TLSv1...	95	Application Data
27	10.997751	192.168.18.70	18.97.36.70	TLSv1...	306	[TCP Previous segment not captured] , Application Data
28	11.345755	18.97.36.70	192.168.18.70	TLSv1...	185	Application Data
32	12.779621	172.64.148.235	192.168.18.70	TLSv1...	90	Application Data
34	12.780202	192.168.18.70	172.64.148.235	TLSv1...	94	Application Data
41	14.563593	192.168.18.70	142.250.193.14	QUIC	1292	Initial, DCID=f57dde696ecedc25, PKN: 2, PING, PADDING, PING, PADDING, CRYPTO, PING, PING, CRYPTO,
48	14.580347	142.250.193.14	192.168.18.70	QUIC	1292	Initial, SCID=f57dde696ecedc25, PKN: 3, CRYPTO, PADDING
67	14.947828	192.168.18.70	192.178.158.84	QUIC	1292	Initial, DCID=4c2537cd4ebd6835, PKN: 1, CRYPTO, CRYPTO, PADDING, CRYPTO, CRYPTO, PING, CRYPTO, PI
68	14.947863	192.168.18.70	192.178.158.84	QUIC	1292	Initial, DCID=4c2537cd4ebd6835, PKN: 2, PING, CRYPTO, CRYPTO, PADDING, CRYPTO, CRYPTO, PING, PADD
75	14.963423	192.178.158.84	192.168.18.70	QUIC	1292	Initial, SCID=ec2537cd4ebd6835, PKN: 3, CRYPTO, PADDING
256	20.071997	192.168.18.70	142.251.157.4	QUIC	1292	Initial, DCID=d9c6f307fe994eeb, PKN: 1, PING, PING, CRYPTO, PING, PING, CRYPTO, CRYPTO, PING, PIN
257	20.072024	192.168.18.70	142.251.157.4	QUIC	1292	Initial, DCID=d9c6f307fe994eeb, PKN: 2, CRYPTO, CRYPTO, CRYPTO, PING, CRYPTO, PING, PING
261	20.109342	142.251.157.4	192.168.18.70	QUIC	1292	Initial, SCID=f9c6f307fe994eeb, PKN: 3, CRYPTO, PADDING
296	23.193248	192.168.18.70	192.178.134.95	QUIC	1292	Initial, DCID=dd90d44859a35c13, PKN: 2, CRYPTO, PING, PING, CRYPTO, CRYPTO, PING
301	23.208207	192.178.134.95	192.168.18.70	QUIC	1292	Initial, SCID=fd90d44859a35c13, PKN: 3, CRYPTO, PADDING
329	24.169676	192.168.18.70	65.1.18.34	QUIC	1242	Initial, DCID=008bd67db57a3fb7, PKN: 0, CRYPTO, PADDING

Packet 27 Details:

- Frame 6: Packet, 624 bytes on wire (4992 bits), 624 bytes captured (4992 bits) on interf...
- Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
 - Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
 - Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)
 - Type: IPv4 (0x0800)
 - [Stream index: 0]
- Internet Protocol Version 4, Src: 192.168.18.70, Dst: 104.18.32.47
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x02 (DSCP: CS0, ECN: ECT(0))
 - Total Length: 610
 - Identification: 0x0000 (0)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: TCP (6)
 - Header Checksum: 0xdd64 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.18.70
 - Destination Address: 104.18.32.47
 - [Stream index: 0]
- Transmission Control Protocol, Src Port: 51531, Dst Port: 443, Seq: 6941, Ack: 1, Len: 5

Raw Packet Data:

```
0000 94 25 33 c7 7c ab 3e e2 dc a2 8c ac 08 00 45 02  .%3.]>.....E-
0010 02 62 00 00 40 00 40 06 dd 64 c0 a8 12 46 68 12  .b.:@.d...Fh-
0020 20 2f c9 4b 01 bb 18 d2 6b 1f 66 c6 4f ec 80 18  />K....k.f.0...
0030 49 3c 9b 0c 00 00 01 01 08 0a 92 01 e7 b1 69 02  I<.....i...
0040 a8 30 c5 05 17 f5 cf 77 02 6b 56 e9 e2 95 b5 59  .0.....w.kV....Y
0050 58 83 d7 eb 40 f0 61 ec 76 fe 4e fc 56 b3 d9 10  X...@.a.v.N.V...
0060 fe 8d 58 f4 ee fd 6d a3 3a 02 3c 36 91 5b 10 40  .X...m...<6.[.@
0070 26 76 51 6b 51 8f 35 9a b6 51 9b b8 c8 8d 09 eb  &vQkQ.5..Q.....
0080 5c 9b 41 7b 8a 13 6f 4f 0c ce 64 a3 7b 50 f1 0d  \A{.o0..d.{P...
0090 23 39 24 43 ff e0 b6 49 2d af df 09 cf 5c e9 b9  #9$C...I....\..
00a0 4b ad 07 35 12 42 e3 41 f9 12 11 a7 fb 44 8a 20  K.5.B.A.....D..
00b0 df 70 b2 de c4 0a dc 84 e4 dc 52 e4 49 ca 00 97  .p.....R.I...
00c0 61 6b 23 bb 88 9b 83 29 60 8c 88 46 b5 bf 57 b0  ak#.....)..F.W-
00d0 69 07 89 9f f4 c8 ac 81 b9 8c 31 45 af 8a f6 44  i.....~1E...D
00e0 39 08 c9 ba fd 1a 91 7e a2 be f5 4b 88 04 6d 60  9.....~.K..m'
00f0 86 e8 09 9d c6 0d 7c 12 3f 51 80 d3 04 5a f0 f6  ....|..?Q...Z..
0100 96 14 b1 54 f2 a1 bc da 97 b7 df a4 07 30 83 3c  ..T.....0<...
0110 bd 02 51 c1 81 46 26 20 e2 9c 1e ee cf f1 57 02  ..Q.F&.....W-
0120 6f b8 6d bf cc 05 b0 a5 53 27 1c 84 dd 07 a7 ae  o.m....S'.....
0130 fa b7 d2 6d c8 93 51 ca e4 76 1c 8b fc b2 cc 1e  ...m..Q..v.....
0140 89 c6 b4 5e 2b 76 6e 4e 22 b2 4f 49 08 c3 65 27  ...^+vNn...0I..e'
0150 55 24 52 59 2e bc 82 97 e8 ce 2c 55 f3 8e fb e7  USRY.....,U....
0160 63 22 74 d9 11 8e 4a 42 24 75 23 ab 7f e6 89 13  c"t...JB $u#....
0170 de af 64 b2 19 e6 36 0c a5 05 10 c2 4f 49 54 bb  .d...6...u.OIT..
0180 f2 3e 63 fd c5 ad 41 e5 5d d3 dc a8 f8 76 81 f4  >c...A. ....v...
```

TLS ANALYSIS

⚡ UDP (User Datagram Protocol) Analysis

The **UDP** filter displayed connectionless communication. Most UDP packets were related to DNS requests, which require fast communication without establishing a connection. This analysis helped in understanding the difference between TCP and UDP communication.

The image shows a Wireshark packet capture analysis of UDP traffic. The top pane displays a list of 20 UDP packets, all originating from 211.850588 and destined for 192.168.18.70. The middle pane shows the details of the selected packet (No. 12319), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response). The bottom pane displays the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
12319	211.850588	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12320	211.850589	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12321	211.850590	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12322	211.852314	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12323	211.852316	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12324	211.852316	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12325	211.852318	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12326	211.852319	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12327	211.852320	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12328	211.852321	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12329	211.852323	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12330	211.852324	142.251.150.119	192.168.18.70	UDP	562	443 → 65395 Len=520
12331	211.852325	142.251.150.119	192.168.18.70	UDP	228	443 → 65395 Len=186
12332	211.852489	192.168.18.70	142.251.150.119	UDP	81	65395 → 443 Len=39
12333	211.854434	142.251.150.119	192.168.18.70	UDP	1286	443 → 65395 Len=1244
12334	211.854436	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12335	211.854437	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12336	211.854439	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12337	211.854439	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12338	211.854440	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250
12339	211.854441	142.251.150.119	192.168.18.70	UDP	1292	443 → 65395 Len=1250

> Frame 3540: Packet, 247 bytes on wire (1976 bits), 247 bytes captured (1976 bits) on interface
> Ethernet II, Src: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab), Dst: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)
> Internet Protocol Version 4, Src: 192.168.18.1, Dst: 192.168.18.70
> User Datagram Protocol, Src Port: 53, Dst Port: 46818
> Domain Name System (response)

```
0000 3e e2 dc a2 8c ac 94 25 33 c7 7c ab 08 00 45 00 >...% 3...E.  
0010 00 e9 75 25 40 00 40 11 1f 47 c0 a8 12 01 c0 a8 ..u@.@.G.....  
0020 12 46 00 35 b6 e2 00 d5 a9 d7 2b fd 81 80 00 01 -F-5...+.....  
0030 00 0b 00 00 00 00 01 69 05 79 74 69 6d 67 03 63 .....i yting c  
0040 6f 6d 00 00 01 00 01 c0 0c 00 01 00 01 00 00 01 om.....  
0050 08 00 04 ac fd 9e 77 c0 0c 00 01 00 01 00 00 01 ...J}w.....  
0060 08 00 04 4a 7d 18 77 c0 0c 00 01 00 01 00 00 01 ...J}w.....  
0070 08 00 04 4a 7d 44 77 c0 0c 00 01 00 01 00 00 01 ...J}w.....  
0080 08 00 04 4a 7d c8 77 c0 0c 00 01 00 01 00 00 01 ...J}w.....  
0090 08 00 04 8e fa 04 77 c0 0c 00 01 00 01 00 00 01 ...w.....  
00a0 08 00 04 ac d9 c2 77 c0 0c 00 01 00 01 00 00 01 ...w.....  
00b0 08 00 04 40 e9 aa 77 c0 0c 00 01 00 01 00 00 01 ...@w.....  
00c0 08 00 04 8e fb 0a 77 c0 0c 00 01 00 01 00 00 01 ...w.....  
00d0 08 00 04 8e fb 0c 77 c0 0c 00 01 00 01 00 00 01 ...w.....  
00e0 08 00 04 4a 7d 82 77 c0 0c 00 01 00 01 00 00 01 ...J}w.....  
00f0 08 00 04 ac fd 76 77 .....vw
```

User Datagram Protocol: Protocol
Packets: 14589 · Displayed: 9828 (67.4%)
Profile: Default

UDP ANALYSIS

TCP SYN Packet Analysis

The filter `tcp.flags.syn==1 && tcp.flags.ack==0` was used to display only the initial SYN packets that start a TCP connection. During the analysis, SYN packets from the client to the server were observed along with some retransmissions. These retransmissions indicated repeated connection attempts when a response was delayed or unavailable. Based on the captured traffic, this behavior was identified as normal connection establishment rather than a confirmed cyberattack. This analysis helped in understanding how cybersecurity professionals examine TCP flags when investigating suspicious network activity.

The image shows a Wireshark packet capture analysis. The top pane displays a list of captured packets, with packet 1007 selected. The middle pane shows the details of packet 1007, which is a TCP SYN packet from 192.168.18.70 to 216.239.34.157. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
737	44.547129	192.168.18.70	163.70.146.60	TCP	78	53535 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3824929803 TSecr=0 SACK_PERM
741	44.564280	163.70.146.60	192.168.18.70	TCP	74	443 → 53535 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1392 SACK_PERM TSval=1402857571 TSecr=3824
1007	45.635013	192.168.18.70	216.239.34.157	TCP	78	51542 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3513018420 TSecr=0 SACK_PERM

Packet Details:

- Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
- Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
- Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)
- Type: IPv4 (0x0800)
- [Stream index: 0]
- Internet Protocol Version 4, Src: 192.168.18.70, Dst: 216.239.34.157
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 64
- Identification: 0x0000 (0)
- > 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: TCP (6)
- Header Checksum: 0x6c3d [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 192.168.18.70
- Destination Address: 216.239.34.157
- [Stream index: 26]
- Transmission Control Protocol, Src Port: 51542, Dst Port: 443, Seq: 0, Len: 0

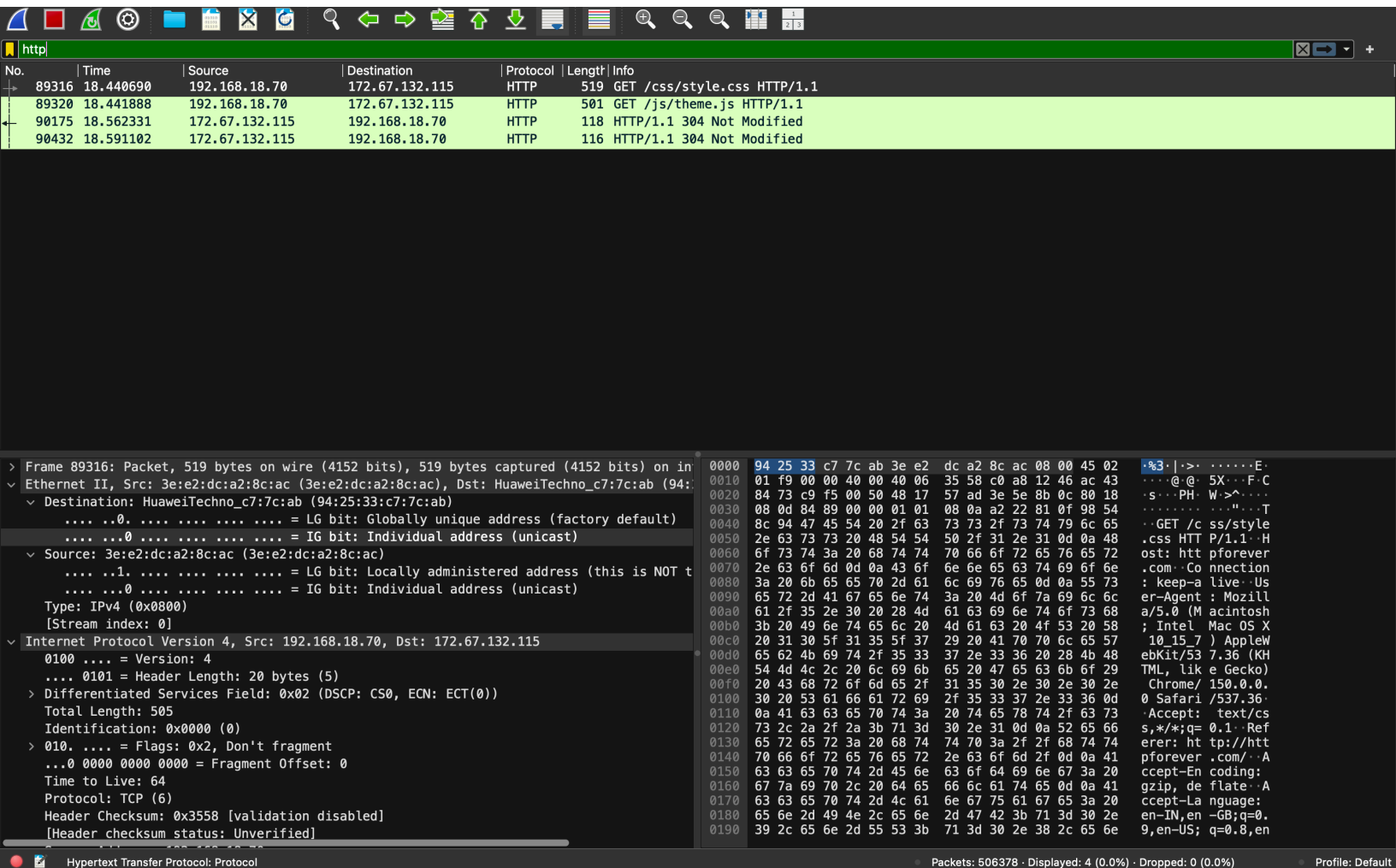
Raw Data:

```
0000 94 25 33 c7 7c ab 3e e2 dc a2 8c ac 08 00 45 00  .%3.|.>.....E.
0010 00 40 00 00 40 00 00 06 6c 3d c0 a8 12 46 d8 ef  .@.@.@.l=...F..
0020 22 9d c9 56 01 bb a7 71 46 d5 00 00 00 b0 c2    ".V..q F.....
0030 ff ff 74 cf 00 00 02 04 05 b4 01 03 03 06 01 01  .t.....
0040 08 0a d1 64 68 34 00 00 00 00 04 02 00 00      ...dh4.....
```

tcp.flags.syn==1 && tcp.flags.ack==0 ANALYSIS

HTTP (Hypertext Transfer Protocol) Analysis

Very little or no HTTP traffic was observed because most websites automatically use HTTPS. This highlighted the increasing adoption of secure communication on the internet. HTTP traffic can still be analyzed when accessing websites that do not use encryption.



The image shows a Wireshark network traffic analysis interface. The top pane displays a list of packets, with packet 89316 selected. The middle pane shows the packet details for Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
89316	18.440690	192.168.18.70	172.67.132.115	HTTP	519	GET /css/style.css HTTP/1.1
89320	18.441888	192.168.18.70	172.67.132.115	HTTP	501	GET /js/theme.js HTTP/1.1
90175	18.562331	172.67.132.115	192.168.18.70	HTTP	118	HTTP/1.1 304 Not Modified
90432	18.591102	172.67.132.115	192.168.18.70	HTTP	116	HTTP/1.1 304 Not Modified

Frame 89316: Packet, 519 bytes on wire (4152 bits), 519 bytes captured (4152 bits) on interface 0
Ethernet II, Src: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac), Dst: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
Destination: HuaweiTechno_c7:7c:ab (94:25:33:c7:7c:ab)
... .. = LG bit: Globally unique address (factory default)
... .. = IG bit: Individual address (unicast)
Source: 3e:e2:dc:a2:8c:ac (3e:e2:dc:a2:8c:ac)
... .. = LG bit: Locally administered address (this is NOT t
... .. = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 192.168.18.70, Dst: 172.67.132.115
0100 = Version: 4
... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x02 (DSCP: CS0, ECN: ECT(0))
Total Length: 505
Identification: 0x0000 (0)
> 010. = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x3558 [validation disabled]
[Header checksum status: Unverified]

0000 94 25 33 c7 7c ab 3e e2 dc a2 8c ac 08 00 45 02
0010 01 f9 00 00 40 00 40 06 35 58 c0 a8 12 46 ac 43
0020 84 73 c9 f5 00 50 48 17 57 ad 3e 5e 8b 0c 80 18
0030 08 0d 84 89 00 00 01 01 08 0a a2 22 81 0f 98 54
0040 8c 94 47 45 54 20 2f 63 73 73 2f 73 74 79 6c 65
0050 2e 63 73 73 20 48 54 54 50 2f 31 2e 31 0d 0a 48
0060 6f 73 74 3a 20 68 74 74 70 66 6f 72 65 76 65 72
0070 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e
0080 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 73
0090 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c
00a0 61 2f 35 2e 30 20 28 4d 61 63 69 6e 74 6f 73 68
00b0 3b 20 49 6e 74 65 6c 20 4d 61 63 20 4f 53 20 58
00c0 20 31 30 5f 31 35 5f 37 29 20 41 70 70 6c 65 57
00d0 65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48
00e0 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29
00f0 20 43 68 72 6f 6d 65 2f 31 35 30 2e 30 2e 30 2e
0100 30 20 53 61 66 61 72 69 2f 35 33 37 2e 33 36 0d
0110 0a 41 63 63 65 70 74 3a 20 74 65 78 74 2f 63 73
0120 73 2c 2a 2f 2a 3b 71 3d 30 2e 31 0d 0a 52 65 66
0130 65 72 65 72 3a 20 68 74 74 70 3a 2f 2f 68 74 74
0140 70 66 6f 72 65 76 65 72 2e 63 6f 6d 2f 0d 0a 41
0150 63 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20
0160 67 7a 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 41
0170 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a 20
0180 65 6e 2d 49 4e 2c 65 6e 2d 47 42 3b 71 3d 30 2e
0190 39 2c 65 6e 2d 55 53 3b 71 3d 30 2e 38 2c 65 6e

...@. 5X...F.C
...PH. W>^...
... ..T
..GET /c ss/style
.css HTT P/1.1..H
ost: htt pforever
.com..Co nnection
: keep-a live..Us
er-Agent : Mozill
a/5.0 (M acintosh
; Intel Mac OS X
10_15_7) AppleW
ebKit/537.36 (KH
TML, lik e Gecko)
Chrome/150.0.0.
0 Safari/537.36
Accept: text/cs
s,*/*;q=0.1 Ref
erer: ht tp://htt
pforever .com/.A
ccept-En coding:
gzip, de flate..A
ccept-La nguage:
en-IN,en -GB;q=0.
9,en-US; q=0.8,en

Hypertext Transfer Protocol: Protocol
Packets: 506378 · Displayed: 4 (0.0%) · Dropped: 0 (0.0%) · Profile: Default

HTTP ANALYSIS

5. Overall Observation

The packet analysis demonstrated how different network protocols work together to enable communication over the internet. By examining packet details such as **Source IP Address, Destination IP Address, Port Numbers, Protocol, Packet Length, Time, and Information**, I gained practical knowledge of network communication and learned the methodology used in network forensic investigations. Although the captured traffic mainly represented normal network activity, the analysis provided a strong foundation for understanding how suspicious traffic can be investigated using packet-level evidence.

6. Suspicious Activity Investigation

Suspicious activity investigation is an important phase of network forensics because it helps identify unusual communication that may indicate a potential security threat. In this project, the focus was on understanding the investigation process using Wireshark and learning how cybersecurity professionals analyze packet-level evidence. Although the captured live traffic represented normal network communication, the investigation methodology for identifying suspicious behavior was studied and practiced.

Investigation Approach

- Captured live network traffic using Wireshark.
- Applied display filters such as **DNS, TCP, UDP, ICMP, TLS, HTTP**, and **tcp.flags.syn==1 && tcp.flags.ack==0**.
- Examined packet details, including **Source IP Address, Destination IP Address, Port Numbers, Protocol, Packet Length, Time, and TCP Flags**.
- Compared the observed communication with normal network behavior to understand how unusual patterns can be recognized.

Indicators of Suspicious Activity

During a forensic investigation, the following behaviors are commonly examined to identify potential security threats:

- A large number of **SYN packets** from the same source IP within a short period.
- One source IP attempting to connect to **multiple destination ports**, which may indicate a **port scan**.
- Repeated **TCP retransmissions** or failed connection attempts.
- Unusual or excessive **DNS requests** to unknown or suspicious domains.
- Unexpected communication with external or unknown IP addresses.
- Abnormally high network traffic that differs from normal communication patterns.

Analysis and Interpretation

Each suspicious indicator should be analyzed carefully before drawing any conclusion. A single packet is **not sufficient** to classify network activity as malicious. Instead, the investigation should consider the overall communication pattern, packet frequency, TCP flags, source and destination IP addresses, destination ports, timestamps, and protocol behavior. Only after examining all available evidence can an activity be classified as normal or potentially suspicious.

Outcome

This investigation provided a clear understanding of the methodology used to identify suspicious network activity through packet analysis. Although no real attack was observed in the captured live traffic, the project helped develop practical skills in recognizing abnormal communication patterns, collecting digital evidence, and understanding the investigation process followed by cybersecurity professionals during network forensic analysis.

7. Findings

The investigation successfully demonstrated how **Wireshark** can be used to capture, monitor, and analyze live network traffic. By applying different display filters and examining packet-level information, I gained practical knowledge of how network communication takes place and how packet analysis supports digital forensic investigations. The major findings of this project are summarized below:

Successful Live Traffic Capture

- Successfully captured real-time network traffic generated through normal internet activities such as web browsing and the **ping** command.
- Saved the captured traffic in **PCAPNG** format for detailed analysis and future reference.

Protocol Analysis

- Applied and analyzed multiple Wireshark display filters, including **DNS, TCP, UDP, ICMP, TLS, HTTP, and TCP SYN**.
- Understood the purpose and behavior of each protocol and its role in network communication.

Packet-Level Investigation

- Examined important packet details such as **Source IP Address, Destination IP Address, Port Numbers, Protocols, Packet Length, Time, and TCP Flags**.
- Observed how devices establish reliable communication through the TCP three-way handshake and how encrypted communication is protected using TLS.



Understanding Suspicious Activity

- Learned the methodology used to investigate suspicious network behavior by studying communication patterns, retransmissions, TCP flags, and unusual packet characteristics.
- Understood that suspicious activity should always be confirmed through detailed analysis rather than relying on a single packet or filter result.



Digital Evidence Collection

- Collected packet captures and Wireshark screenshots as digital evidence.
- Documented all observations, analysis, and conclusions in a structured forensic investigation report.



Knowledge and Skills Gained

- Developed practical skills in **network packet analysis, protocol investigation, and digital evidence collection.**
- Gained a strong understanding of the workflow followed by cybersecurity professionals during **network forensic investigations.**
- Built a solid foundation for future learning in **Cybersecurity, Digital Forensics, Incident Response, and Security Operations Center (SOC)** roles.

8. Conclusions

This project successfully demonstrated the practical application of **network packet forensics** using **Wireshark**. Throughout the investigation, live network traffic was captured, filtered, and analyzed to understand how devices communicate over a network. By examining protocols such as **DNS, TCP, UDP, ICMP, TLS, and HTTP**, I gained practical knowledge of packet-level analysis and the role of digital evidence in cybersecurity investigations.

Key Outcomes of the Project

- Successfully captured live network traffic using Wireshark.
- Applied different display filters to analyze various network protocols.
- Examined important packet details, including **IP addresses, ports, protocols, packet lengths, timestamps, and TCP flags**.
- Learned how network communication can be monitored and documented through packet analysis.
- Prepared a structured **Forensic Investigation Report** based on the observations and findings.

Understanding Suspicious Packet Investigation

Although my live network capture contained only normal communication, I learned the standard methodology used by cybersecurity professionals to investigate suspicious network activity. If a suspicious **PCAP** file is available from trusted cybersecurity repositories or laboratory environments, it can be opened in Wireshark and examined using appropriate display filters. The investigator analyzes packet details, communication patterns, TCP flags, source and destination IP addresses, ports, and protocol behavior to determine whether the traffic indicates activities such as a port scan, SYN flood, malware communication, or other network attacks. This structured approach helps ensure that conclusions are based on evidence rather than assumptions.

Importance in Cyber Attack Investigation

Network packet forensics plays a vital role in **Cyber Attack Investigation** because every packet captured on a network can serve as digital evidence. By analyzing packet captures, investigators can reconstruct communication between systems, identify suspicious behavior, understand how an incident occurred, and support incident response with technical evidence. These investigations are widely used in **Security Operations Centers (SOC)**, digital forensic laboratories, corporate security teams, and law enforcement agencies to strengthen cybersecurity and investigate cybercrime.

Final Remarks

This project provided a strong foundation in **network packet analysis, digital evidence collection, and forensic investigation techniques**. It improved my understanding of network communication and introduced me to the practical workflow followed by cybersecurity professionals. The knowledge and skills gained through this project will support my future learning in **Cybersecurity, Digital Forensics, Incident Response, and Network Security**, enabling me to investigate network incidents more effectively as I continue developing my career in this field.

9. References

The following resources were used throughout this project to understand network packet analysis, protocol behavior, and the forensic investigation process. These references provided both theoretical concepts and practical guidance for using Wireshark effectively.

PETV134L – Cybernetx: Summer Internship in Cyber Security and Network Intelligence

- Online internship course conducted by **Lovely Professional University (LPU)** during **June–July**.
- The course provided practical guidance on cybersecurity concepts, network intelligence, and the methodology of packet analysis and digital forensic investigation.
- The live sessions and demonstrations served as a valuable foundation for completing this project.

Wireshark Official Documentation

- Used to understand the features of Wireshark, packet capture techniques, display filters, and protocol analysis.
- Helped in learning how to capture and examine live network traffic accurately.

Malware Traffic Analysis

- Studied as a learning resource to understand how suspicious network traffic and cyberattacks are investigated using packet captures.
- Provided examples of real-world forensic investigation techniques.

Netresec PCAP Repository

- Referred to as a trusted source of publicly available PCAP files used for cybersecurity education and forensic analysis.
- Helped in understanding how investigators analyze network traffic captured during security incidents.